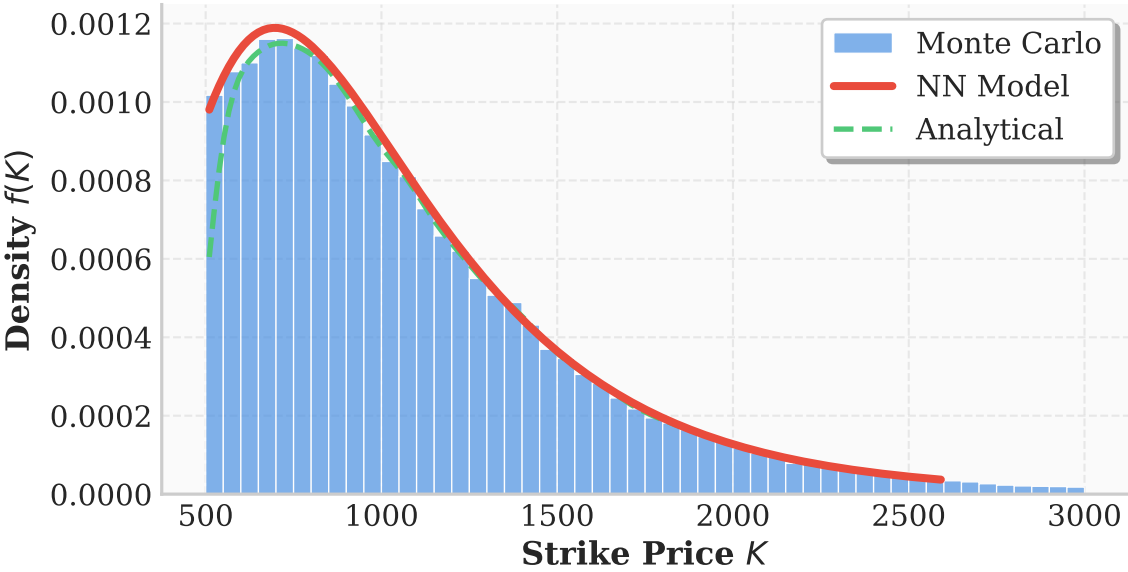


PDF Analysis: Neural Network Synthetic Local Volatility Model

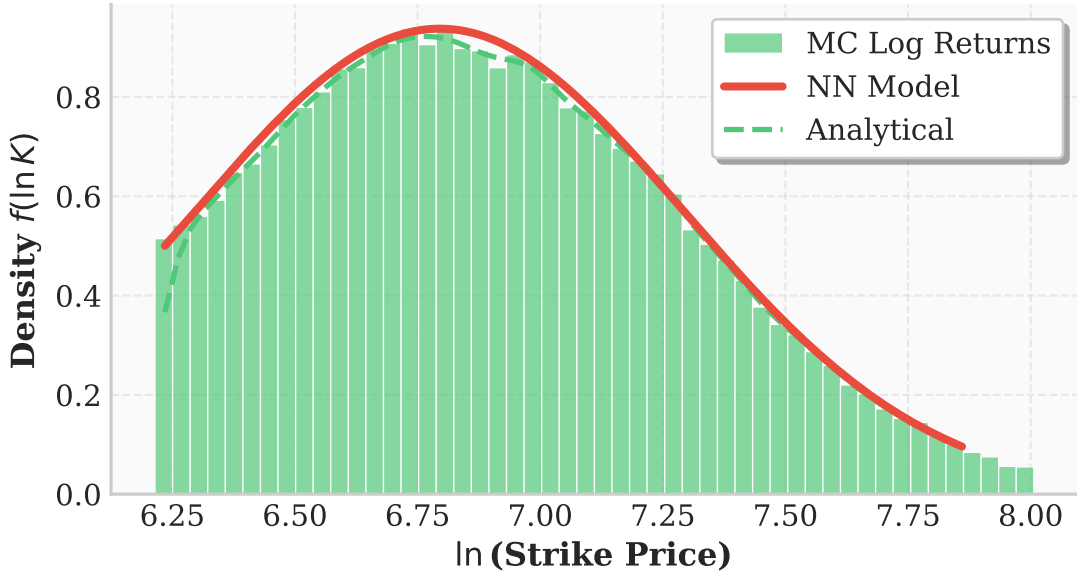
Monte Carlo under learned σ_{NN} : $dS_t = rS_t dt + \sigma_{NN}(t, S_t)S_t dW_t$

MC source: data

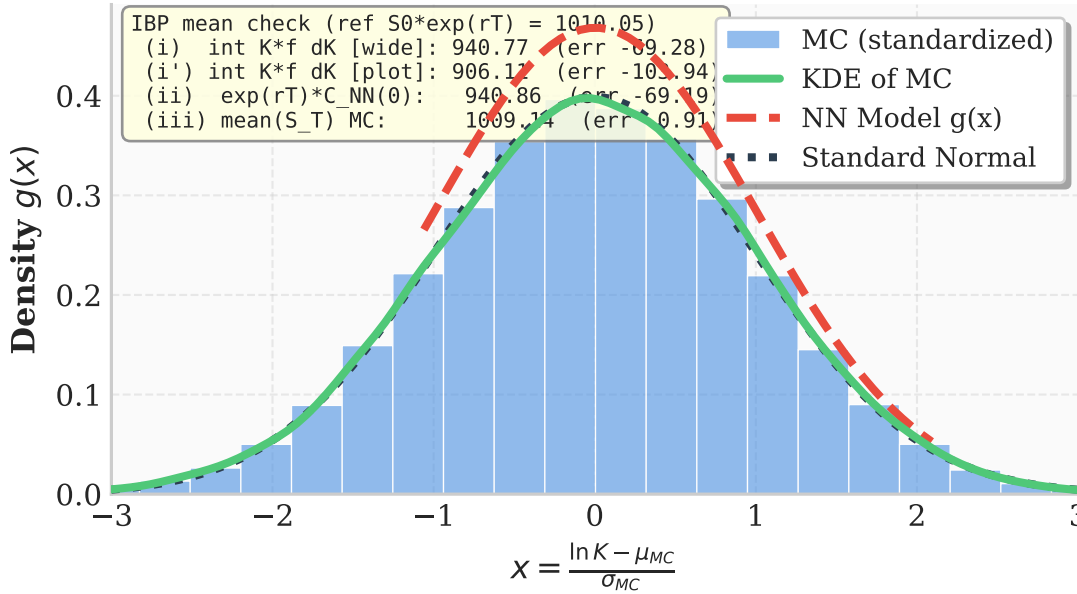
Strike Distribution (T = 0.25)



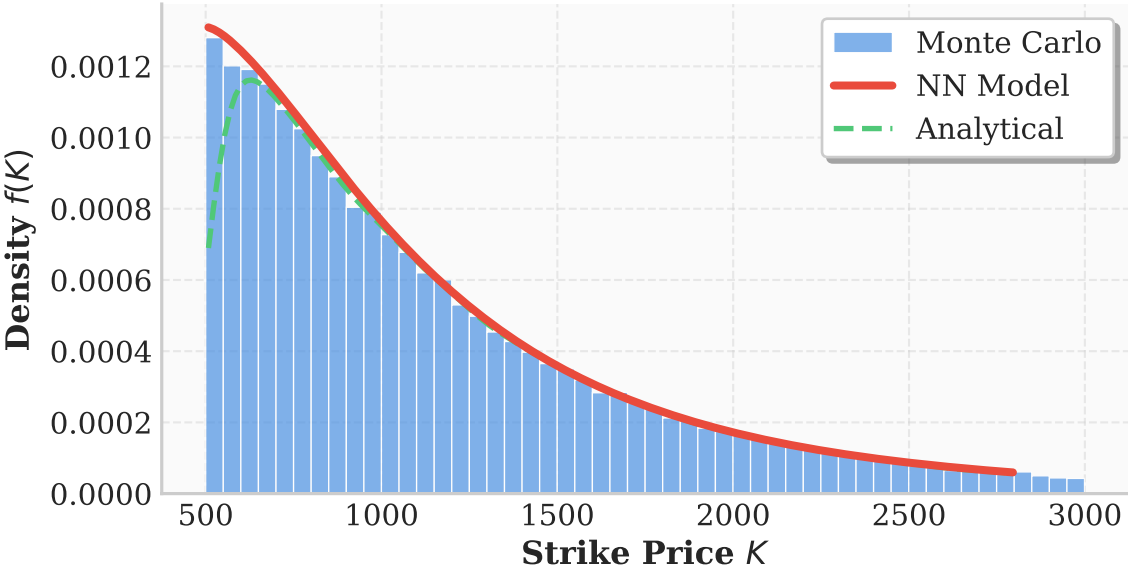
Log-Normal Distribution (T = 0.25)



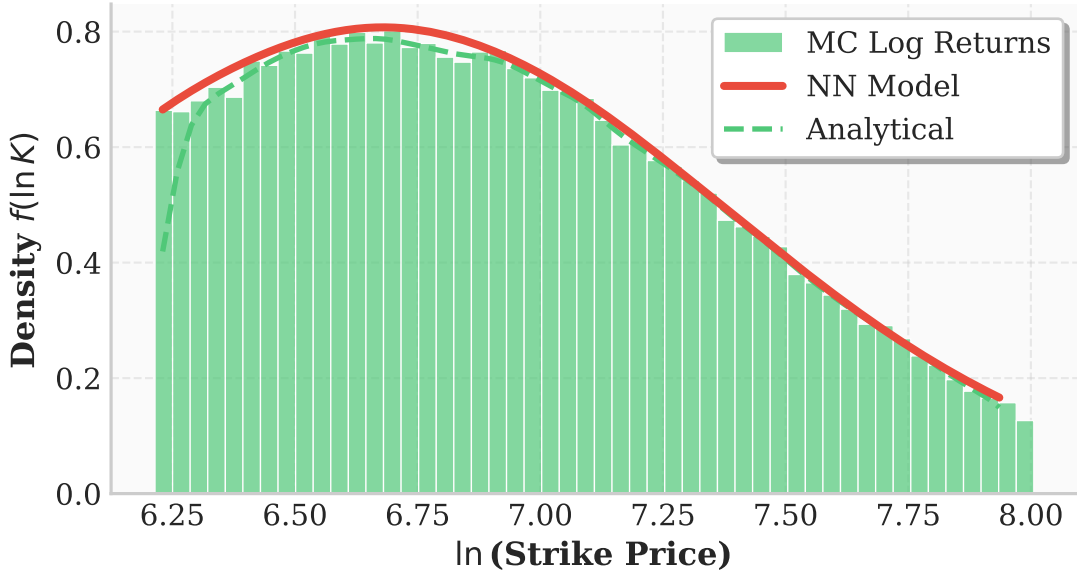
Gaussian Space (T = 0.25)
MC Standardization: $\mu_{MC}=6.79, \sigma_{MC}=0.50 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.89, \sigma_{model}=0.39$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.02, \text{ExKurt}=-0.02$



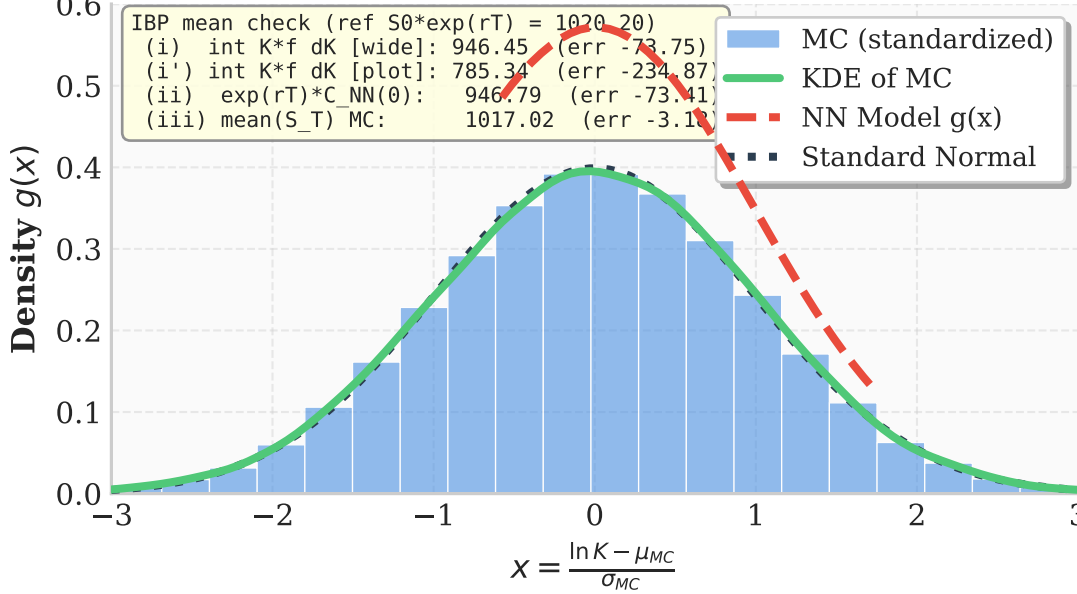
Strike Distribution (T = 0.50)



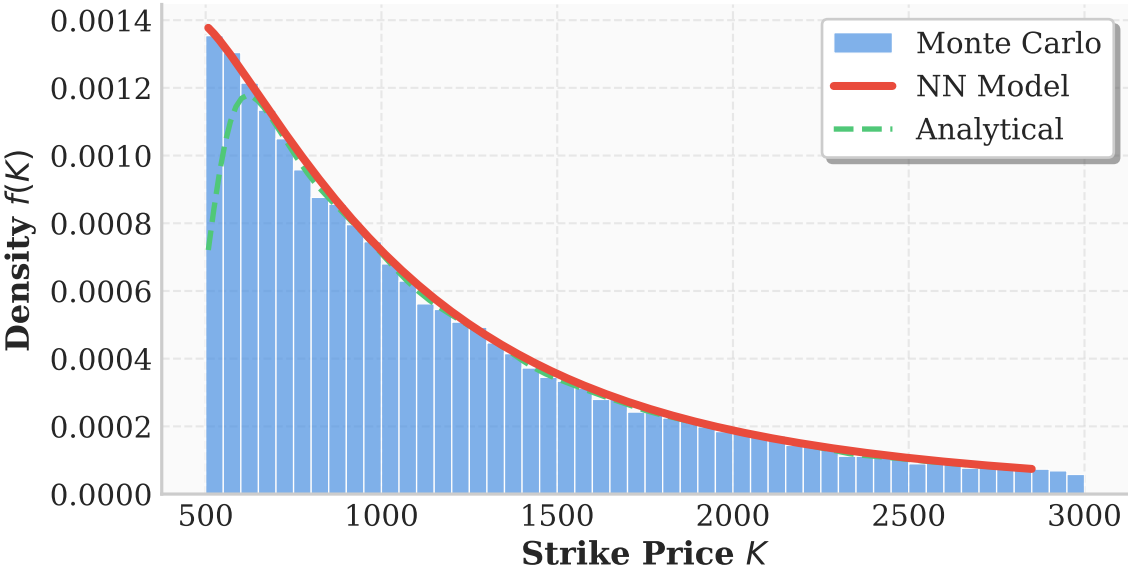
Log-Normal Distribution (T = 0.50)



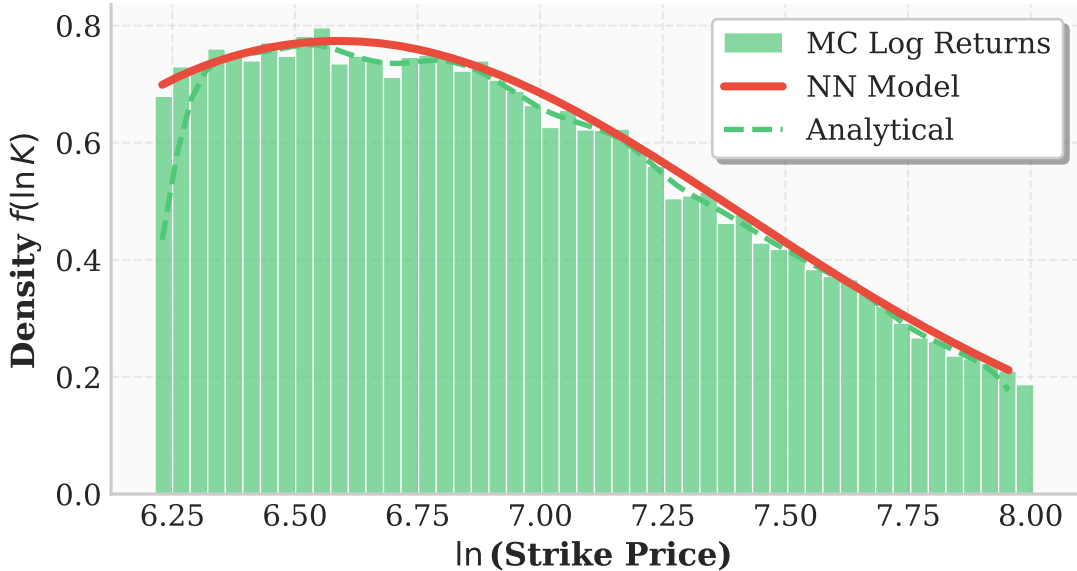
Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.67, \sigma_{MC}=0.71 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.92, \sigma_{model}=0.44$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=-0.02$



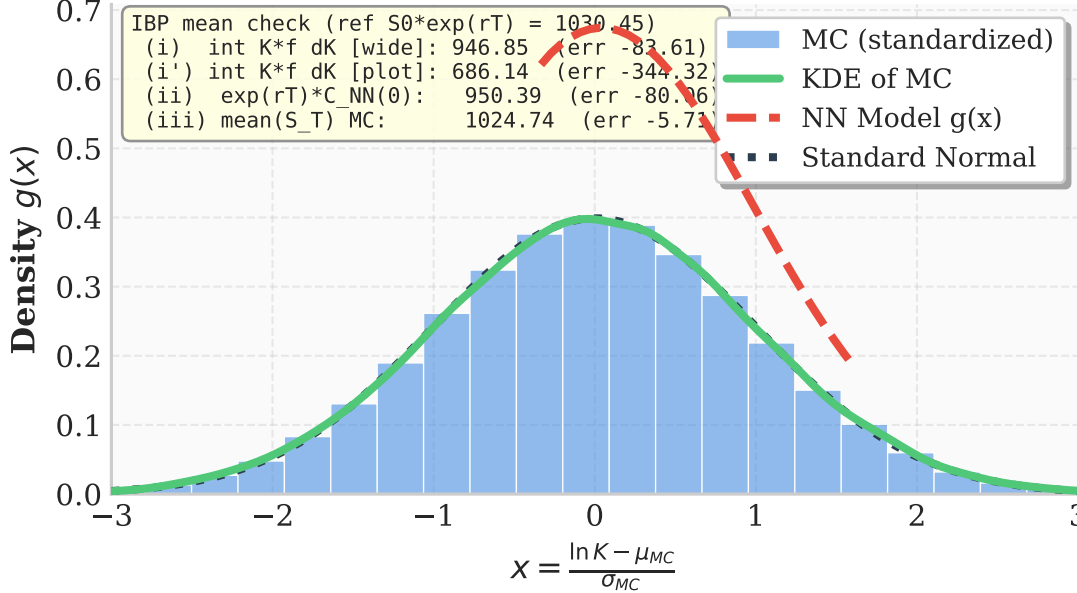
Strike Distribution (T = 0.75)



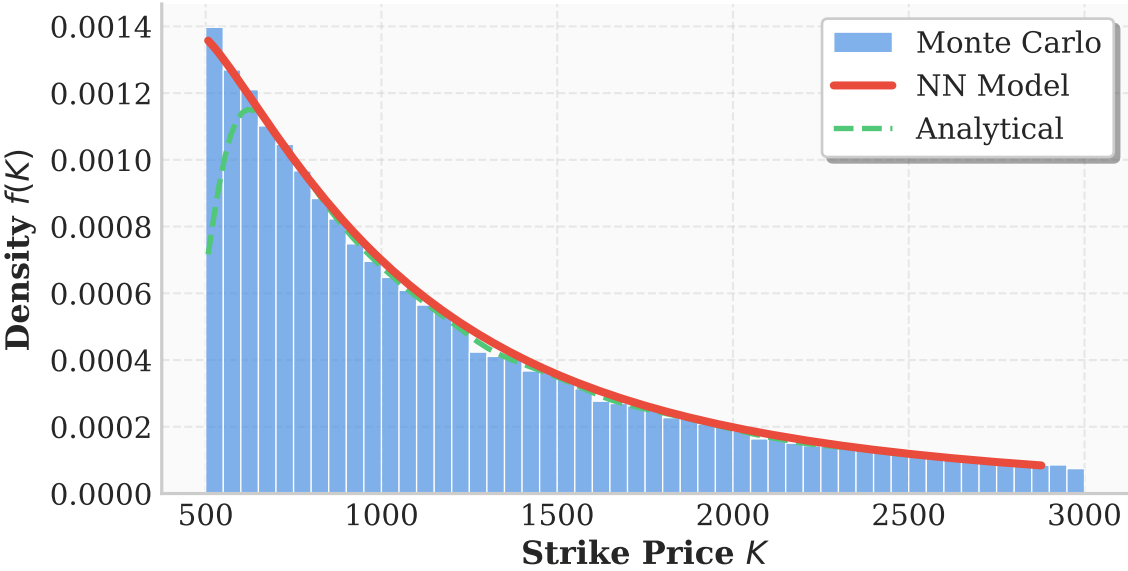
Log-Normal Distribution (T = 0.75)



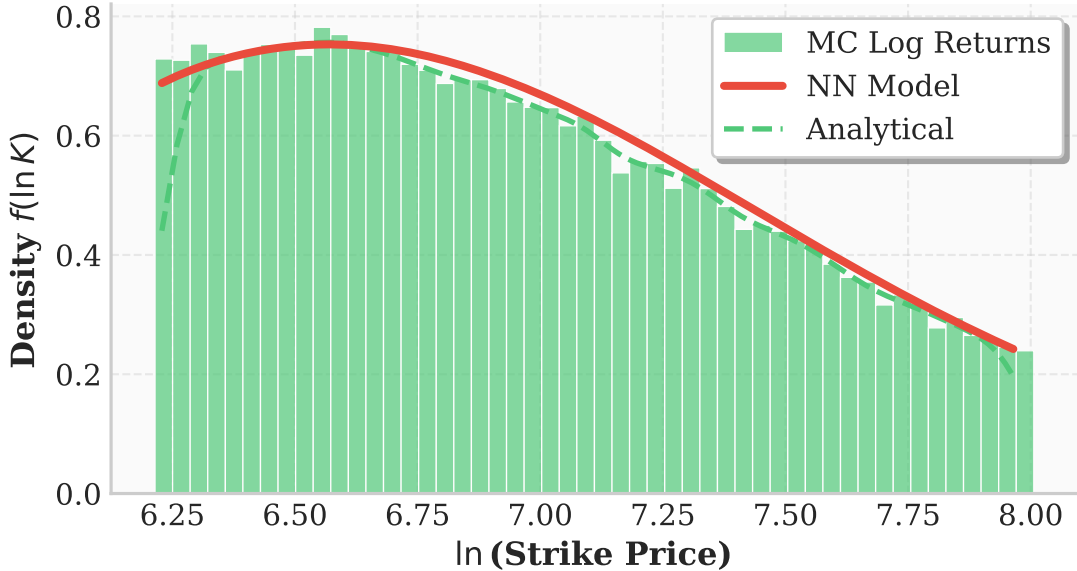
Gaussian Space (T = 0.75)
MC Standardization: $\mu_{MC}=6.55, \sigma_{MC}=0.87 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.94, \sigma_{model}=0.46$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.00, \text{ExKurt}=-0.01$



Strike Distribution (T = 1.00)



Log-Normal Distribution (T = 1.00)



Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.44, \sigma_{MC}=1.01 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.96, \sigma_{model}=0.46$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=0.01$

